



Dog Skin Diseases Detection and Identification Using Convolutional Neural Networks

Anand Upadhyay¹ · Gangashankar Singh¹ · Siddhesh Mhatre¹ · Penil Nadar¹

Received: 1 November 2022 / Accepted: 18 December 2022 / Published online: 4 March 2023
© The Author(s), under exclusive licence to Springer Nature Singapore Pte Ltd 2023

Abstract

When veterinary clinics diagnose dogs with common skin diseases, they often do not have a record of these diseases. The reason for this is that they have to deal with so many different types of animals each day. In addition, this may delay the diagnosis of other non-visual apparent diseases in other patients. Based on supervised learning for classify images, we use the convolutional neural networks to classify images of dog skin diseases at the same level as dermatologists and diagnosis could be made faster and more accurate, potentially saving lives. We can also make software applications and put them on machines like mobile devices and computers. Implementing a system useful to prepare for your pre-examination, you need to identify common skin diseases in dogs. To achieve these goals, processes are used to manipulate images, and Convolutional Neural Network (CNN) is considered to be an ideal approach. The veterinary clinic will benefit from having such a system in place for its day-to-day operations.

Keywords Dog skin diseases classification · Authentication · Convolutional neural network · Supervised learning model · Classification · Deep learning

Introduction

The most common animal we see everywhere is the dog, whether at home, in the police station, in the military, or even at dog shows [2]. Dogs are trained for everything and everywhere due to this, and they are more chances of getting skin disease. Nowadays, due to global warming or due to some climate change, there are many diseases. Due to this,

the dogs are affected. The dogs are very sensitive to climate they must be handled carefully. Due to climate change or unusual dogs getting affected their skin does not suit the climate, so they get different types of skin disease. These skin diseases can be natural or due to some insects, so it is necessary to identify these skin diseases with the help of CNN algorithm. Convolutional neural networks (CNNs) have recently undergone improvements that have improved their performance in these cutting-edge image identification tasks and systems [6]. If a dog falls sick, we have to go to the doctor and ask him and he tells us to take a blood test, urine test, etc., to find out the cause or disease. It is estimated that between 20 and 75% of veterinarian consultations for dogs and cats in clinics and hospitals are due to skin issues [7]. Skin experts can distinguish dog skin types with less effort and expense using a high-performance classifier. In the literature, many machine learning and deep learning-based approaches for classifying skin disorders have been presented. To identify wound photos into several classifications, such as Ring Worm, Fleas, and Mange, we created an ensemble Deep Convolutional Neural Network-based classifier in this study [9]. Following a veterinarian's clinical examination, blood samples (10 mL from the jugular vein for serology and 40 L from the tip of the ear on filter paper

This article is part of the topical collection “Advances in Computational Intelligence for Artificial Intelligence, Machine Learning, Internet of Things and Data Analytics” guest edited by S. Meenakshi Sundaram, Young Lee, and Gururaj K S.

✉ Anand Upadhyay
anandhari6@gmail.com
Gangashankar Singh
sg.indra@nic.in
Siddhesh Mhatre
siddeshmhatre1649@gmail.com
Penil Nadar
peniljones450@gmail.com

¹ Department of IT, Thakur College of Science and Commerce, Thakur Village, Kandivali (East), Mumbai, Maharashtra 400101, India

for DNA extraction) were taken (IsoCodeTM, Schleicher & Schuell, Keene, NH) [8]. But now with the help of our research, we can find out the disease and help them to cure it as soon as possible. Our research will help to identify what type of skin disease is it. We have classified three causes of skin disease. For classification of skin diseases that we have taken is Ringworm, Fleas, and Mange. We have taken data from hospitals and from Internet sources in such a way that just by taking photos, we can identify the diseases. Our research will help to get the output in less time and we have not to wait for days for the result. This technique will help to find the disease and cure it as soon as possible. We have used 240 images for training and 59 images for validation. With this algorithm in the future, we can implement it on camera or in drones to help street dogs or in society in which dogs are not able to carry or dogs which are helpless, so we can capture the images and help them by sending medicine or by sending doctors or helpers to help the dogs on spot and prevent the disease to spread from one to another. This research can be used by the government to help dogs for preventing skin disease and make dogs skin disease free.

Literature Review

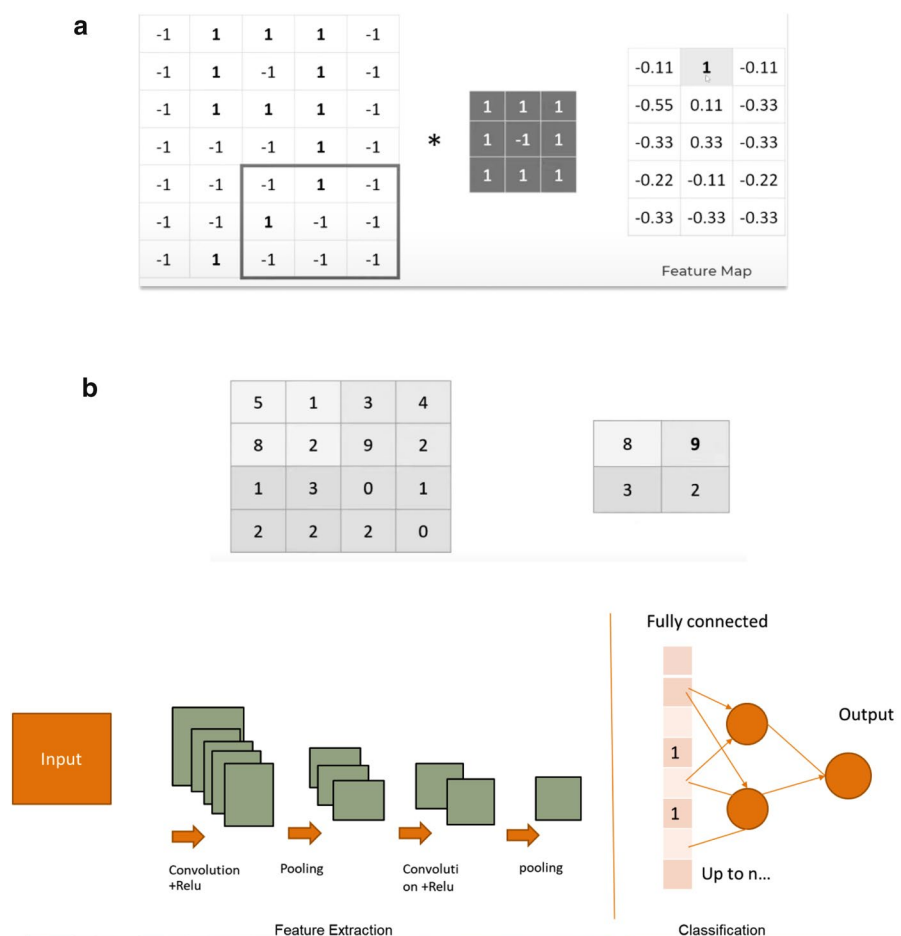
Beau Gray in his paper has developed a system in which we can identify very frequent dog skin diseases for identification purpose and he has created numerical results that can be used for veterinary clinics [4]. To reliably quantify the likelihood of infection in seropositive animals, frequent random surveys using non-serological tests are required due to the specificity change of serology according to prevalence (Universidade Federal do Piauí, in his paper). Creating new mathematical models is crucial if you want to get the genuine prevalence from seroprevalence [8]. Bay L, Bames in his findings shows that healthy hip and knee skin have comparable microbial compositions, although the epidermal and dermal compartments have different compositions and functions. The dermal compartment, which was a particular subset of the epidermal microbiome, often featured a more homogenous microbial makeup among people. All of these results point to less variation in the human cutaneous microbiome than previously thought. The dermal communities may more precisely represent the host's genetic or immunological condition rather than being a byproduct of the host's external environment, which is an unanticipated conclusion that makes a significant addition to our knowledge of the human skin microbiota in health and illness [1]. As cutting-edge skin lesion classifiers, CNNs perform exceptionally well. Unfortunately, it is challenging to evaluate different categorization systems, since some of them rely on non-public datasets for testing and/or training, which makes repeatability challenging. Future papers should include thorough

disclosures about the training methods they utilized and use publicly accessible benchmarks to enable comparison [3]. The skin identification approach in this study based on skin-and-non-skin binary-classifier is more efficient and resilient when compared to previous skin detection methods based on special color space (Shen in his paper). We discover that the skin detection approach presented in this work has a significant anti-interference capability [10]. With the help of the image processing technique, he can identify the models and convolutional neural network. He has trained the model and used the main core with Central Processing Unit (CPU) and with approach of a convolutional neural network. He has used GPU to increase the speed and training of the model. This system will help the veterinary to solve the problem and to help the dogs as soon as possible. He has not ignored the efficiency and accuracy and train it with full perfection and with concern. In our research, we implemented CNN with Adam optimizer with three classes and we got 82% accuracy and 0.45 loss. We have trained the model in such a way to achieve maximum accuracy and efficiency in identifying skin diseases.

Algorithm

CNN has two main convolution and pooling types of layers. CNN's fundamental building piece is the convolution layer. The majority of the network's computational burden is carried by it. This layer creates a dot product between two matrices, where one matrix represents the kernel—a collection of parameters that may be learned—and the other matrix represents the constrained area of the receptive field shown in Fig. 1.1. Although the kernel is deeper than an image, it is spatially smaller. This indicates that the kernel height and width, if a picture has three (RGB) channels, will be spatially modest, but the depth will go up to all three channels. This gives us output after passing through many activation layers in CNN, and scanning is done in machine language. The machine identifies pixels in any image or collects information by scanning a group of the pixel through a filter. The binary-classifier is the foundation for skin area detection. Due to the 50% accuracy of random classification, it is quite simple to create a binary-classifier with high accuracy. A CNN model extracts the picture's feature vector, and the classifier is then used to categorize the image and output the probability of skin or non-skin to identify skin [10] and collects information by scanning a group of pixels through a filter. This filter can be in a different size like 3×3 filter, which has a random value which can be seen in Fig. 1.1. When this filter scans or convolutes over any photo, it combines the filter's random number with the photo's pixel count to form a new set. And all this helps to find the quality of

Fig. 1 1 Illustration of the convolution process. 2 Pooling process illustration. Illustration of convolutional neural network



the photo. In the same way, all other possible property is scanned and puts into CNN. This is fed into the CNN's output in a fully connected layer. Its end output helps to classify the photo to get the answer normalization or legalization technique is performed.

Data are reduced using mean or max-pooling in the pooling layer. By obtaining a summary statistic from the surrounding outputs, the pooling layer substitutes for the network's output at certain points. As a result, the representation's spatial dimension is reduced, which reduces the amount of computation and weights needed. Every slice of the representation is independently handled for the pooling procedure. The pooling process illustration is shown in Fig. 1.2.

While the primary pooling determines the average value, the max-pooling will choose the highest value. All the neurons from the preceding activation layer will be connected by this layer. The logistic regression technique may be modified to implement SoftMax activation to classify more than two groups.

Methodology

This research we collected data in the form of color pictures from clinical and from Internet for the condition of Mange, ringworm, and fleas are dog skin diseases. The distribution of training images and validation images was 75% and 25%, resulting in 240 pictures for training and 59 images for validation and 240 for training. The mobile device's picture size was (1920, 1080), making it too big to be taught. TensorFlow 2.10.0 was used to crop and resize each skin picture to (180, 180) [5]. The resolution of skin image inputs is modified to 180×180 pixels based on Fig. 2 for the CNN have three layers. Convolutional neural networks do a lot of processing on the pictures prior to feeding the inputs into the fully connected layer, even though they also contain a fully connected neural network component at the conclusion of the process, as shown in Fig. 1.2. Their efforts are directed on shrinking the photos and transforming them into a format that retains

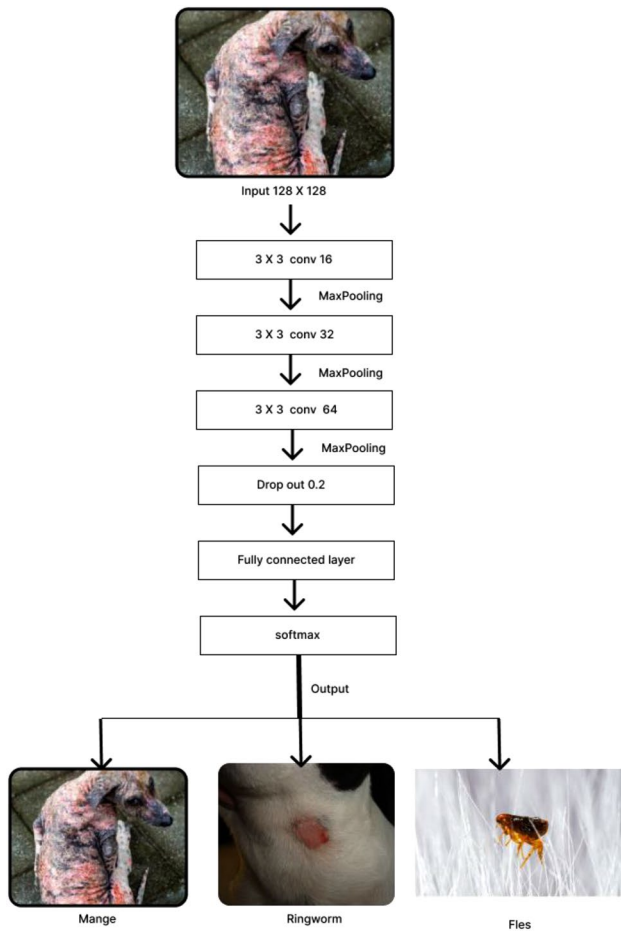


Fig. 2 System model for skin diseases

the elements needed to distinguish between the various classes while also being much smaller and simpler to process [11]. Convolutions and pooling make up the majority of this effort, drastically reducing the amount of inputs delivered to the fully connected layer. The picture is distorted by applying three-by-three filters to each hidden layer, with 16, 32, and 64 output channels on each layer, correspondingly. ReLU activation and Max pooling are used in the activation procedure at each layer, notably in this model, where it serves as the default activation function. The ReLU function it uses arithmetic computation. We use the SoftMax function related to the cross-entropy function. Figure 2 shows how the size of the picture is reduced because of Max pooling. For optimization, we use Adam optimization. The Adam optimizer combines concepts from many previous optimizers. Adam uses a previous gradient average that decays exponentially, just as the momentum optimizer. The flattening procedure will then reduce the image's three-dimensional characteristics to one dimension. The next step will be to apply the SoftMax activation algorithm to divide the three categories

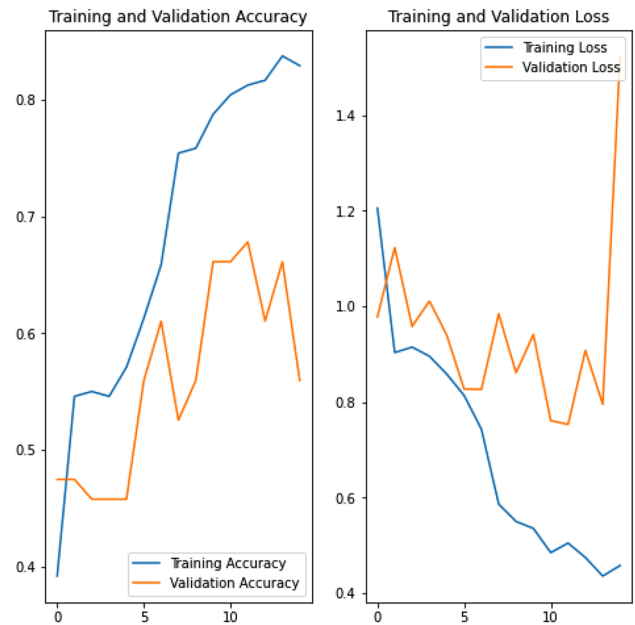


Fig. 3 Graph illustrating accuracy

of skin conditions into mange, fleas, and ringworm. The output is in the form of image with skin diseases label.

Results

In this research, the training model employed 240 training photos and 59 validation images and 240 for training. The images obtained from the clinic consisted of three classes' mange, Ringworm, and fleas. All images are trained by the CNN algorithm with the Adam optimizer method, an optimizer having a learning rate of 0.001, and implement categorical cross-entropy. Adam optimizers generally produce better results, take less time to compute, and require fewer parameters to tune than any other optimization algorithm. We did 15 iterations (epoch), and we can compare accuracy and loss performance in Fig. 3. Adam optimizer gives better performance.

The accuracy and loss graph shows in Fig. 3 a lot of spikes, which suggests as much. This model achieves 82% Training Accuracy with 55% Validation Accuracy and 46% loss. While the accuracy of the system model that applies Adam optimization, training and validation data are less, and the system false also starts to decline at each iteration. The proposed CNN model can classify fleas, ringworm, and mange with good performance and gives low error. In Fig. 4, the result is in the form of a color image with a skin diseases class label.

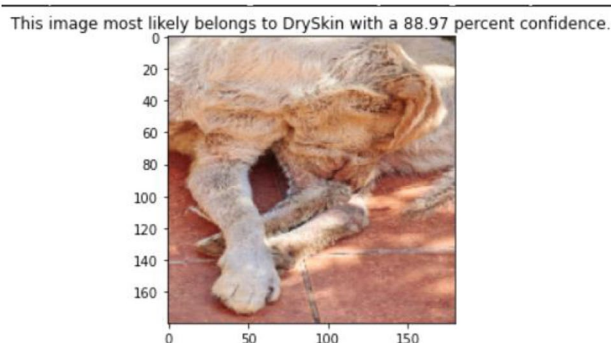


Fig. 4 Result

Conclusion

This research developed an automatic identification of skin disease. The work done in this thesis shows that multiple models can achieve higher accuracy than what human experts can achieve. The diagnosis is based on the presence of mange, fleas, ringworms, and mange skin analyzing digital images. The CNN model works with three hidden layers, three fully connected layers, and soft-max activation. The three hidden layers utilize 3×3 filter sizes with 16, 32, and 64 channel outputs in succession. CNN with Adam optimizer can give better output of classifying the dog skin diseases data with 82% accuracy and a loss of 45%. The method shows that the suggested model is very useful to use as a current tool for hospitals' professionals in making a diagnosis of skin diseases in dogs. Systems to categorize the numerous forms of skin diseases and other type of skin diseases could be created with more study.

Funding This research received no external funding.

Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

Ethical approval All applicable international, national, and/or institutional guidelines for the care and use of animals were followed.

Informed consent Not applicable.

References

1. Bay L, Barnes CJ, Fritz BG, Thorsen J, Restrup MEM, Rasmussen L, et al. Universal dermal microbiome in human skin. *MBio*. 2020;11:e02945-e3019.
2. Bray EE, Otto CM, Udell MAR, Hall NJ, Johnston AM, MacLean EL. Enhancing the selection and performance of working dogs. *Front Vet Sci*. 2021. <https://doi.org/10.3389/fvets.2021.644431>.
3. Brinker TJ, Hekler A, Utikal JS, Grabe N, Schadendorf D, Klode J, Berking C, Steeb T, Enk AH, Von Kalle C. Skin cancer classification using convolutional neural networks: systematic review. *J Med Internet Res*. 2018;20(10): e11936.
4. Habal BGM, Tiong PES, Pasatiempo JR, Balen MJ, Amarga MR, Juco L. Dog skin disease recognition using image segmentation and GPU enhanced convolutional neural network. In: 2021 IEEE 13th international conference on humanoid, nanotechnology, information technology, communication and control, environment and management (HNICEM). 2021.
5. Hwang S, Shin HK, Park JM, Kwon B, Kang MG. Classification of dog skin diseases using deep learning with images captured from multispectral imaging device. *Mol Cell Toxicol*. 2022. <https://doi.org/10.1007/s13273-022-00249-7>.
6. Khan S, Rahmani H, Shah SAA, Bennamoun M, Medioni G, Dickinson S. A guide to convolutional neural networks for computer vision. Morgan Claypool [Online]. 2018. Available from: <https://ieeexplore.ieee.org/document/8295029>
7. Narita FB, Scardoeli B, von Ancken ADCB, de Paula Coelho C. Homeopathic treatment in dog dermatopathy. *Braz J Dev*. 2020;6(1):2209–15.
8. Reis LE, Coura-Vital W, Roatt BM, Bouillet LÉ, Ker HG, de Brito RC, de Melo RD, Carneiro M, Giunchetti RC, Marques MJ, Carneiro CM. Molecular diagnosis of canine visceral leishmaniasis: a comparative study of three methods using skin and spleen from dogs with natural *Leishmania infantum* infection. *Vet Parasitol*. 2013;197(3–4):498–503.
9. Rostami B, Anisuzzaman DM, Wang C, Gopalakrishnan S, Niezgoda J, Yu Z. Multiclass wound image classification using an ensemble deep CNN-based classifier. *Comput Biol Med*. 2021;134: 104536.
10. Shen X, Zhang J, Yan C, Zhou H. An automatic diagnosis method of facial acne vulgaris based on convolutional neural network. *Sci Rep*. 2018;8(1):1–10.
11. Sumit Saha. A comprehensive guide to convolutional neural networks—the ELI5 way. Available from: <https://towardsdatascience.com/a-comprehensive-guide-to-convolutional-neural-networks-the-eli5-way-3bd2b1164a53>. Accessed 09 May 2022.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.